Varsha Balaji

Chicago, IL | +14022372373 | vbala7@uic.edu
linkedin.com/in/varsha-balaji | github.com/B-Varsha | b-varsha.github.io

Education

University of Illinois at Chicago — Aug. 2024 – May 2026
Masters of Science in Computer Science — *GPA: 4.00/4.00*
Coursework: *Data Science, Computer Algorithms, Neural Networks, Natural Language Processing, AI, Big Data Mining*

SSN College of Engineering — Sep. 2020 – July 2024
Bachelor of Engineering in Computer Science and Engineering — *GPA: 3.65/4.00*

Experience

Machine Learning Engineer Intern — Jun 2025 – Aug 2025
Vosyn AI — *Chicago, IL*

- Engineered and fine-tuned a custom high-fidelity zero-shot voice cloning Text-to-Speech (TTS) deep learning model for multilingual audio synthesis using semantic-to-acoustic tokenization.
- Designed a parallelized, GPU-accelerated data pipeline deployed on GCP (Vertex AI) that processed 800K+ audio files using dynamic in-memory batching and concurrent uploads, achieving 5× throughput improvement.
- Integrated Vertex AI Experiment Tracking and executed distributed model training via Custom Jobs, improving experiment reproducibility by 98% and reducing training iteration setup time by 75%, streamlining the workflow.
- Implemented automated checkpointing, error recovery to ensure fault-tolerant batch inference at scale.

Data Science Intern — July. 2022 – Aug 2022
Synergy Maritime — *Chennai, India*

- Engineered ML-powered virtual sensors for "Scrubber Management," applying advanced regression techniques to improve accuracy from 60% to 85%, enhancing environmental compliance for over 30 maritime vessels.
- Streamlined data pipelines, reducing preprocessing time by 30%, and implemented hyperparameter tuning to optimize model performance, achieving a 25% accuracy improvement. Presented findings to stakeholders, contributing to data-driven decision-making.

Projects

AI-Stylish Fashion Agent - GKE Hackathon | *RAG, GKE, Cloud Run, Streamlit* — Aug. 2025 – Sep 2025

- Engineered and deployed a containerized microservice on **Google Kubernetes Engine (GKE)** to power a vibe-based fashion recommendation system using **Gemini 2.5 Flash** and a **Pinecone vector database**.
- Designed a data ingestion pipeline with **gRPC** for real-time embedding generation, enabling a live **Retrieval-Augmented Generation (RAG)** system.

Progressive In-Context Alignment for LLMs | *LLMs, Gen AI, HuggingFace, NLP* — Feb. 2025 – May 2025

- Implemented the Progressive In-Context Alignment (PICA) inference method, leveraging **Mistral** and **LLaMA** models to improve instruction alignment by encoding few-shot demonstrations.
- Designed a multi-stage inference pipeline to extract **In-Context Learning (ICL) vectors** and inject them to steer targeted, query-only generation.
- Evaluated PICA performance against zero-shot and few-shot baselines using **BLEU, ROUGE-L, and cosine similarity**, confirming consistent performance gains across models.

Facial Antispoofing | *Transformer Networks, Flask* — Sept. 2023 – Apr. 2024

- Developed an advanced facial antispoofing system using **Vision Transformers (ViTs)** and **Exploratory Data Analysis (EDA)** to clean and optimize training data.
- Created proprietary **SPAN (Cross-Layer Relation)** and **MSWF (Feature Fusion)** techniques, which significantly improved model accuracy and robustness for real-time use.
- Designed and deployed a **Flask-based web application** with **WebRTC** for real-time image capture and dynamic client-server interactions.

Publications

- P. Ravisankar, V. Balaji, et al. Deep learning-based renal stone detection: A comprehensive research study and performance analysis. Applied Computer Systems. | Paper

Technical Skills

Programming Languages: Python, Java, C++, C, SQL, JavaScript, HTML, CSS, LaTeX
Frameworks & Libraries: TensorFlow, PyTorch, Flask, ReactJS, Node.js, Bootstrap, Apache Spark, Apache Kafka, Pandas, Pinecone DB, NumPy, OpenCV, Matplotlib, Seaborn,Streamlit, Scikit-learn
Cloud & Development Tools: Google Cloud Platform (GCP), Vertex AI, AWS, Docker, Git, MLflow, Kubernetes, GKE
Software Engineering: Agile (Scrum), CI/CD, Data Structures & Algorithms (DSA) , Object-Oriented Programming (OOP), SOLID Principles, System Design
Other: Agentic AI, Data Cleaning, Data Warehousing, Exploratory Data Analysis (EDA), Ensemble Learning, Explainable AI, FastAPI